Project Summary

Demonstration of technology through project

This project demonstrates the integration of ENVIRONMENTAL MONITORING TECHNOLOGY using IoT, sensors, and cloud-based data analysis. The system measures temperature, humidity, air quality, and light intensity, providing real-time alerts through LEDs and an OLED display, while also sending data to the cloud for remote monitoring via ThingSpeak.

Connections to Advanced Technologies:

Internet of Things (IoT) Integration

- System built on IoT connectivity using ESP32 microcontroller.
- Multiple sensors collect environmental data in real time.
- ESP32's Wi-Fi enables communication with ThingSpeak Cloud for remote data logging and visualization.
- Demonstrates continuous monitoring and intelligent decision-making for smart cities, home automation, and industrial safety systems.

Embedded Systems and Sensor Interfacing

- Combines hardware and software into an integrated embedded system.
- Sensors used:
  - DHT22 → Temperature & humidity
  - MQ-2 → Air quality
  - LDR → Light intensity
- Data displayed on 0.96" OLED module.

Components Required

- ESP32 Microcontroller – brain of the system.
- DHT22 Sensor – measures temperature and humidity.
- MQ-2 Sensor – measures air quality or presence of harmful gases.
- LDR (Light Dependent Resistor) – measures light intensity.
- OLED Display (0.96") – shows real-time readings locally.
- LEDs – indicate alert conditions (Temperature, Air Quality, Light).

Circuit Connections

➢ DHT22 (Temperature & Humidity Sensor)

- The DHT22 usually has 4 pins: VCC, Data, NC (not connected), GND.
- VCC → 3.3V (ESP32 3.3V pin)
- Data → GPIO15 (ESP32 digital input pin)
- GND → GND (ESP32 ground)
- NC → Not connected

➢ MQ-2 Gas / Air Quality Sensor

- MQ-2 has 4 pins for analog reading: VCC, GND, DO (digital output), AO (analog output).
- VCC → 5V (ESP32 5V pin)
- GND → GND (ESP32 ground)
- AO → GPIO34 (ESP32 analog input pin for air quality measurement)
- DO → Not used (optional if you want a digital HIGH/LOW alarm)

➢ Potentiometer

- The potentiometer has 3 pins: VCC, SIG (signal/wiper), and GND.
- VCC → 3.3V (ESP32 3.3V pin)
- SIG → GPIO35 (ESP32 analog input pin)
- GND → GND (ESP32 ground)

➢ LEDs

- LEDs are used as indicators. Connect each anode (+) through a 220–330Ω resistor to a digital pin and cathode (−) to GND.
- Temperature LED (TEMP_LED) → GPIO18(ESP32 analog input pin)

- Light LED (LIGHT_LED) → GPIO19(ESP32 analog input pin)
- Air Quality LED (AIR_LED) → GPIO23(ESP32 analog input pin)

➢ **OLED Display (0.96" I2C)**

- OLED uses I2C interface (4 pins): VCC, GND, SDA, SCL
- VCC → 3.3V (ESP32)
- GND → GND (ESP32 ground)
- SDA → GPIO21(ESP32 analog input pin)
- SCL → GPIO22(ESP32 analog input pin)

3.1.2 Software Setup

- Wokwi Platform – used for programming ESP32.
- Libraries required:
  - DHT.h for temperature and humidity sensor
  - Adafruit_SSD1306.h for OLED display
  - WiFi.h and HTTPClient.h for Wi-Fi connection and cloud communication
- ThingSpeak cloud platform to send and visualize data remotely.
- API key is required to upload data fields (temperature, humidity, air quality, light).

3.1.3 Key Learning Outcomes:

- Hands-on experience with IoT devices, sensors, and cloud platforms.
- Understanding data visualization via OLED displays and web platforms.
- Conceptual insight into satellite telemetry, drone navigation, and remote sensing.
- Exposure to entrepreneurial potential in space technology and environmental solutions.

Screenshot of the project

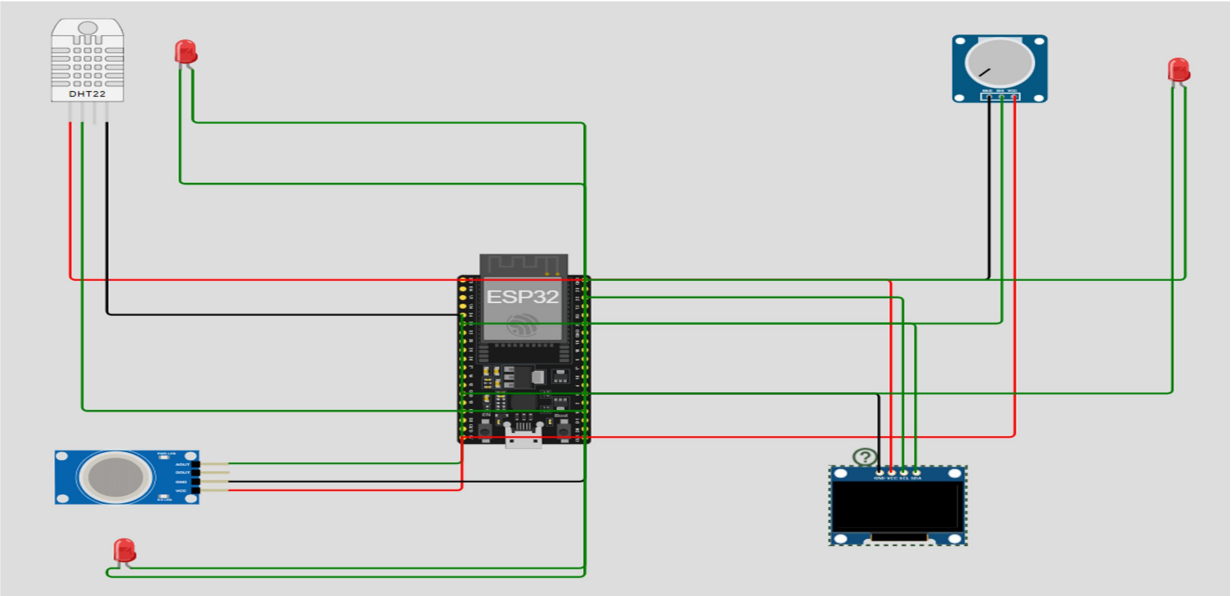

Fig 4.1 Circuit Diagram of IoT-Enabled Environmental Monitoring System Using ESP32

This diagram illustrates an IoT-based environmental monitoring and alert system using an ESP32. It integrates sensors like DHT22 (temperature & humidity), MQ-2 (gas), LDR (light), and ultrasonic sensor (distance), with real-time data shown on an OLED display and sent to ThingSpeak for remote monitoring. LEDs provide visual alerts when environmental parameters exceed predefined thresholds.

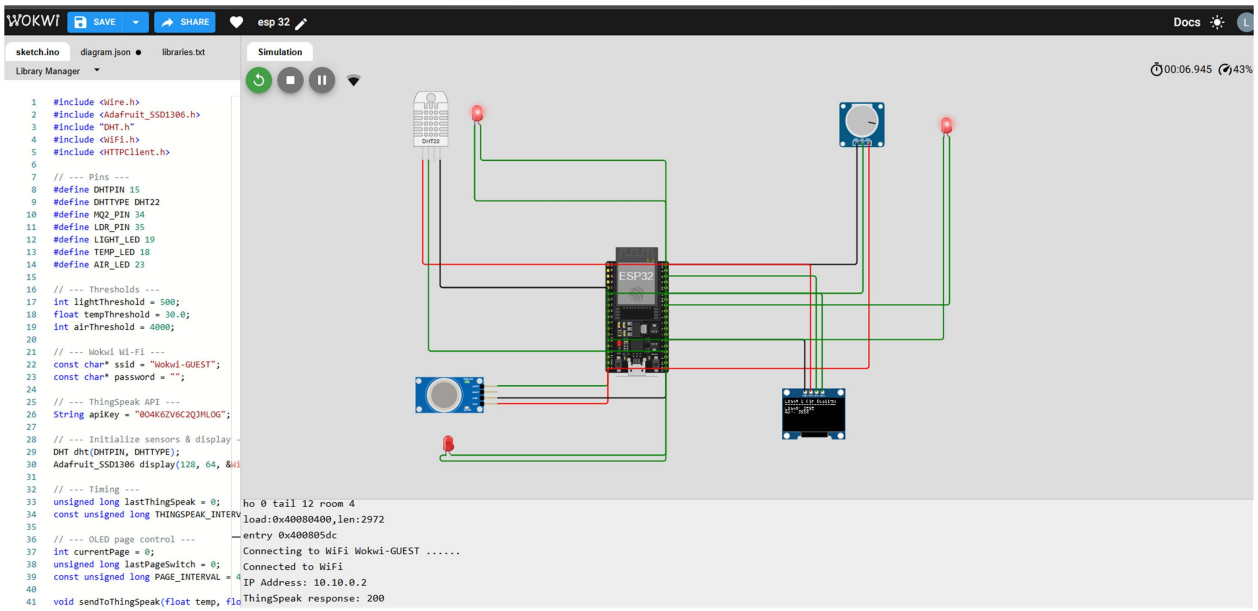

Fig 4.2 Working Circuit Diagram of IoT-Enabled Environmental Monitoring System Using ESP32 on WOKWI

This Wokwi simulation demonstrates an IoT-based Environmental Monitoring and Alert System using the ESP32. It integrates a DHT22 sensor, MQ-2 gas sensor, LDR (via potentiometer), and ultrasonic sensor for real-time data collection. The OLED display shows live readings, while LEDs provide visual alerts. The serial monitor confirms Wi-Fi connectivity and successful data upload to the ThingSpeak cloud platform.

Output obtained on OLED Screen

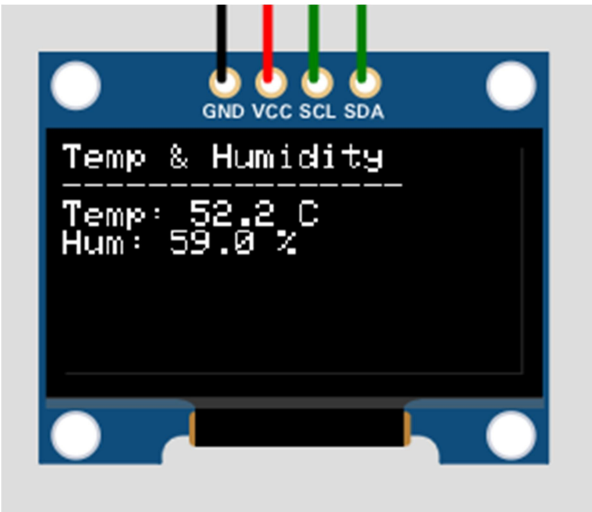

Fig 4.3 Values of Temperature and Humidity

The OLED display shows real-time temperature and humidity readings from the DHT22 sensor, with a temperature of 52.2°C and humidity at 59.0%.

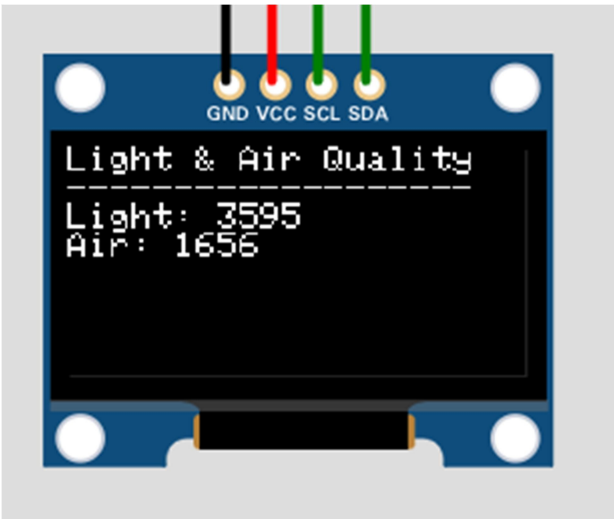

Fig 4.3 Values of Light & Air Quality

The OLED display shows real-time light & air quality readings from the MQ-2 gas sensor, LDR (via potentiometer) with a light of 1656 & air quality of 3595.

Fig 4.3 LED Status & Alerts

The image displays an OLED screen showing the status of various LED indicators: "Light LED: ON," "Temp LED: ON," and "Air LED: OFF." It also shows an alert for temperature, air and light conditions.

Data Uploaded on Thinkspeak

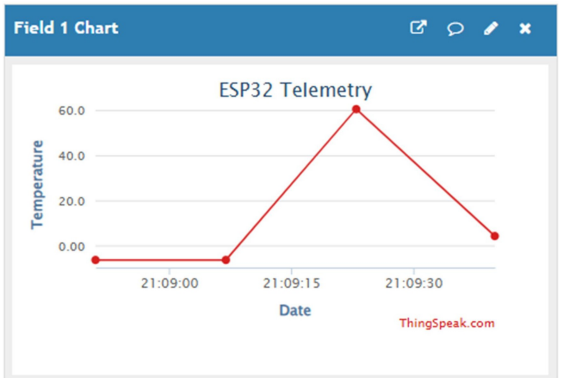

Fig 4.4 Graph of Temperature

The image represents a temperature chart displaying telemetry data over time from an ESP32 device, showing how the temperature changes at different time intervals.

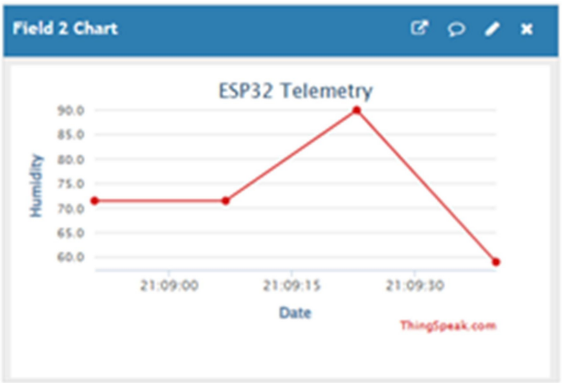

Fig 4.4 Graph of Humidity

The image represents a humidity chart displaying telemetry data over time from an ESP32 device, showing how the humidity changes at different time intervals.

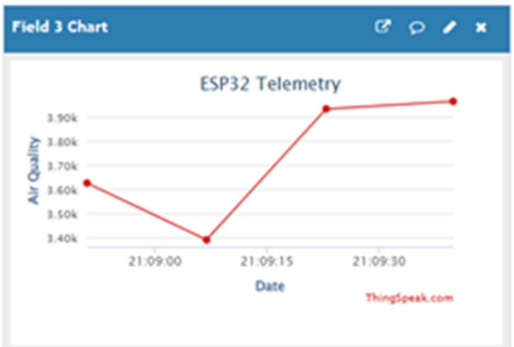

Fig 4.4 Graph of Air Quality

The image represents an air quality chart displaying telemetry data over time from an ESP32 device, showing how air quality changes at different time intervals.

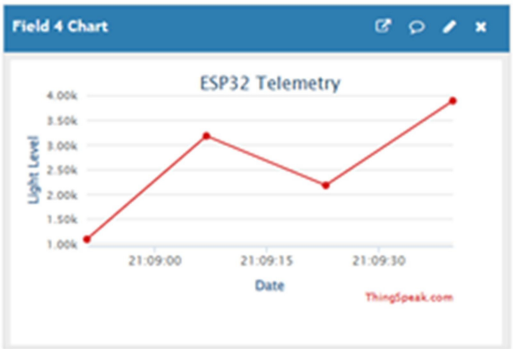

Fig 4.4 Graph of Light Level

The image represents light level chart displaying telemetry data over time from an ESP32 device, showing how light level changes at different time intervals.